

Research interests

Natural language processing, tabular data and knowledge graphs, working towards artificial intelligence for data integration. Particular interest in table representation learning and the problems it raises, such as data scarcity, and in retrieval-augmented generation for industrial use cases such as troubleshooting.

Experience

2023 – now **Doctoral Researcher** · [DTAI, KU Leuven](#) · E2E core lab, [Flanders Make@KU Leuven](#)
Table representation learning and semantic table annotation; knowledge graph construction and ontology engineering. Technical solutions for industrial partners, including retrieval-augmented generation systems for troubleshooting and ontology-driven root cause analysis. Advisor: [Prof. Anastasia Dimou](#).

Education

2023 – now **PhD in Computer Science** · [KU Leuven](#), Belgium
2020–2022 **MSc in IT & Cognition** · [University of Copenhagen](#), Denmark
2014–2018 **BEng in Telecommunication Engineering** · [Shenzhen University](#), China

Publications

† equal contribution. **Bold** marks my name.  paper  slides  poster  project site  data
 DOI

Peer-reviewed conference papers

- [1] I. Dasoulas, S. Vandeveld, J. Jordens, **D. Yang**, X. Duan, A. Bey-Temsamani, J. Vennekens, and A. Dimou. **Can we GLUE it? Extending an Interactive Adhesive Selector with Knowledge Graphs**. In: *International Semantic Web Conference (ISWC 2026)*. 2026.
Accepted, in press · .
- [2] **D. Yang**, I. Dasoulas, and A. Dimou. **Label-Constrained Column Annotation with Language Models and Graph Neural Networks**. In: *IEEE International Conference on Data Engineering (ICDE 2026)*. IEEE, 2026, pp. 2278–2293. 
Research track · **CORE A*** · *To appear* ·  · .
- [3] C. Fleiner, **D. Yang**, S. Vandeveld, and J. Vennekens. **A Domain Ontology for Ishikawa Diagrams to Enhance Root Cause Analysis**. In: *International Semantic Web Conference (ISWC 2025)*. Springer Nature Switzerland, 2025, pp. 258–275. 
Resource track · .
- [4] I. Dasoulas, **D. Yang**, and A. Dimou. **MLSea: A Semantic Layer for Discoverable Machine Learning**. In: *European Semantic Web Conference (ESWC 2024)*. Springer Nature Switzerland, 2024, pp. 178–198. 
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Workshop and challenge papers

- [5] M. Iferroudjene, L. Poggel, A. Schimmenti, **D. Yang**, K. Shivashankar, J.-C. Kalo, and M. Boscariol. **A Wikidata-Based Framework to Measure Cross-Lingual Bias in Multilingual Large Language Models**. In: *Knowledge Graphs and Large Language Models Workshop (KG-LLM) @ LREC 2026*. LREC. European Language Resources Association (ELRA), 2026, pp. 190–209. 
To appear · .
- [6] **D. Yang**, I. Dasoulas, and A. Dimou. **Matching Table Metadata to Knowledge Graphs: A Data Augmentation Perspective**. In: *ELLIS Workshop on Representation Learning and Generative Models for Structured Data 2025*. 2025.
 · .

- [7] I. Dasoulas, **D. Yang**, X. Duan, and A. Dimou. **TorchicTab: Semantic Table Annotation with Wikidata and Language Models**. In: *Semantic Web Challenge on Tabular Data to Knowledge Graph Matching (SemTab) @ ISWC 2023*. Vol. 3557. CEUR Workshop Proceedings. 2023, pp. 21–37. † Dasoulas and Yang contributed equally · Accuracy track, 1st Prize ·  ·  · .
- [8] **D. Yang** and N. Hollenstein. **PLM-AS: Pre-trained Language Models Augmented with Scanpaths for Sentiment Classification**. In: *Northern Lights Deep Learning Workshop (NLDL 2023)* 4 (2023). .

Demonstrations and posters

- [9] I. Dasoulas, **D. Yang**, and A. Dimou. **MLSeascape: Search over Machine Learning Metadata Empowered by Knowledge Graphs**. In: *European Semantic Web Conference (ESWC 2024), Satellite Events*. Springer Nature Switzerland, 2025, pp. 193–196. .

Awards and honours

- 2024 Best Poster Award, E2E cluster · Flanders Make Scientific Conference
- 2023 Accuracy track, **1st Prize** · SemTab 2023 challenge @ ISWC 

Software and datasets

MLSea / MLSO ontology

Semantic layer and ontology for discoverable machine learning metadata.

Ishikawa Diagram Ontology

Domain ontology supporting root cause analysis.

Talks

- 2026 Label-Constrained Column Annotation with Language Models and Graph Neural Networks
ICDE 2026, conference talk · 
- 2023 TorchicTab: Semantic Table Annotation with Wikidata and Language Models
SemTab challenge @ ISWC 2023, challenge talk · 

Teaching

- 2024–25, 2025–26 Web AI
- 2025–26 Programmeertechnieken / Programming Techniques
- 2024–25 Statistiek en databeheer / Statistics and Data Management

Master's thesis supervision

- 2025–26 Ontology Matching with Large Language Models: An Experimental Comparison
- 2025–26 Numerical Table Analysis with Language Models
- 2023–24 Column Qualifier Annotation Using Pretrained Language Model

Availability

Seeking a **research internship** in winter/spring 2026–2027, and **full-time researcher or scientist positions** from July 2027.